

TED (10) 5046

RegNo:.....

Signature.....

Revision 2010

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

BIOSENSORS AND TRANSDUCERS

Maximum Marks:100

MD

Time:3 Hours

PART .A

I. Answer the following questions in One or Two sentences each question carries 2 marks

- 1) Define biosensors?
- 2) Name any four sensors of vision?
- 3) Define pH?
- 4) Give the two examples of optical sensor?
- 5) Give the classification of strain gauge?

5x2=10

PART.B

II. Answer any five of the following. Each question carries 6 marks.

1. Describe any two receptors in human body?
2. Mention the classification of transducers?
3. Describe membrane electrodes?
4. Explain hydrogen electrodes?
5. Explain fiber optic sensor?
6. mention the importance of transducers in biological measurement?
7. Explain thermoelectric p-n junction transducer?

5x6=30

PART.C

Answer one full question from each unit.

Unit - I

- III. a) Explain any two active and passive transducers? 8
b) Write short note on biological sensors? 7

OR

- IV. a) Explain any two sound sensors ? 8
b) Explain any two sensors of smell? 7

Unit – II

- V. a) Explain the working of PCO₂ Electrode with a neat diagram? 8
b) Explain construction of calomel electrode? 7

OR

- VI. a) Explain with Schematic the working of Ag-AgCl electrode? 9
b)) Explain glass electrode for pH measurement ? 6

Unit - III

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| VII. | a) Describe the construction of sensor using ISFET ? | 8 |
| | b) Explain optical sensor? | 7 |

OR

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| VIII. | a) Explain with a diagram the construction of enzyme electrodes? | 8 |
| | b) Explain the different types of fibre optic sensors? | 7 |

Unit - IV

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| IX. | a) Explain LVDT with neat diagram? | 8 |
| | b) Write note on chemical thermometry? | 7 |

OR

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| X. | a) Explain the operation of a displacement transducer? | 8 |
| | b) Explain piezoelectric transducer? | 7 |